

BIOL 343

Applied Bioinformatics

Building pipelines

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Managing computational pipelines

Problems when using many tools:

1. Poor file management

Input/output files end up in unpredictable locations

2. Uninformative file names

Names tend to get loooooong with suffixes like `_new_reallynew_1.txt`

3. Difficult to checkpoint

*Re-running one step sometimes requires re-running **all** the steps*

4. Difficult to take notes

When testing new tools, it's difficult to recall what worked and didn't work (and why)

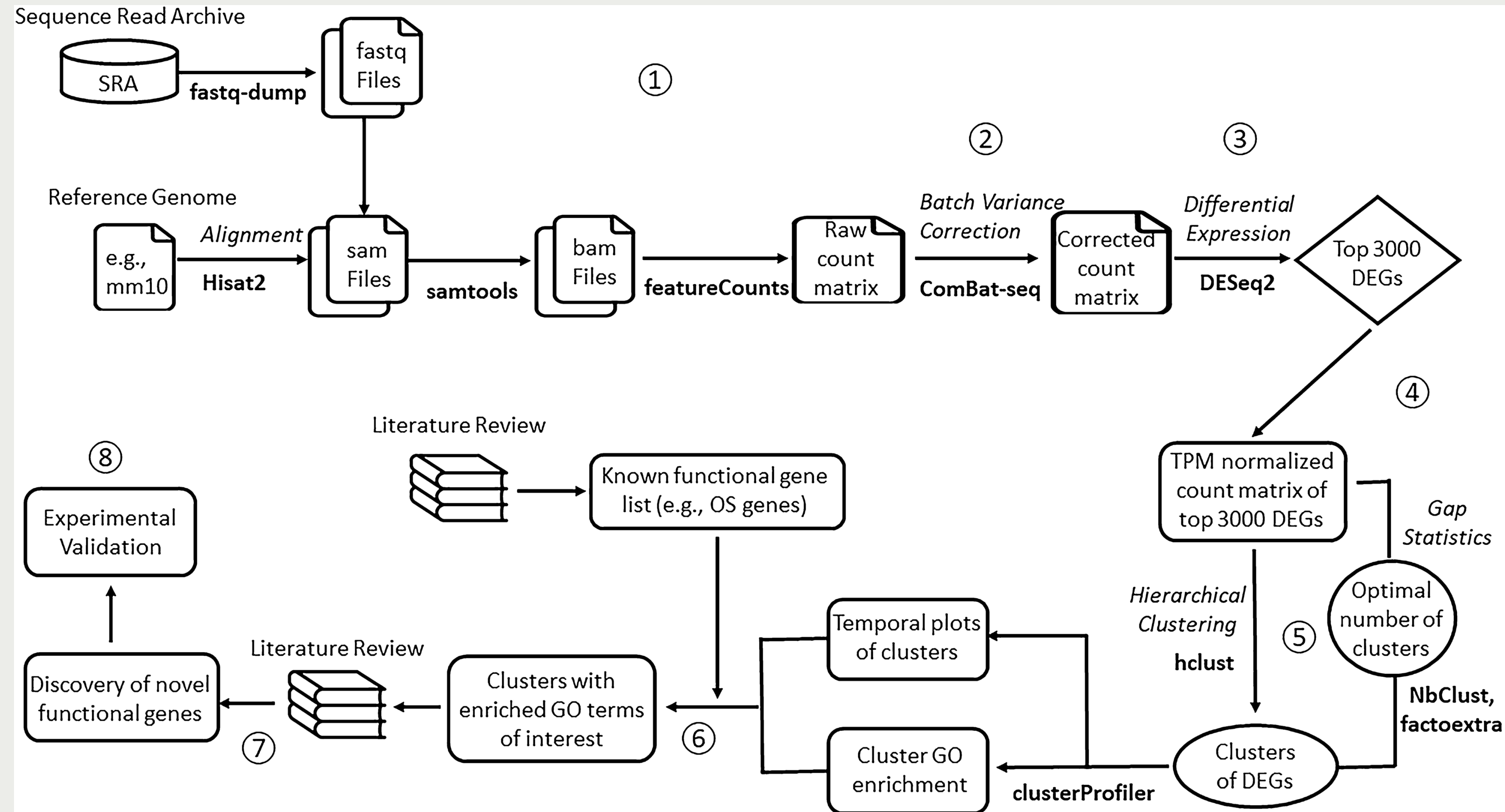


Workflow (pipeline) scripts

Managing computational pipelines

Options for managing a pipeline like this:

- Run one-by-one with CLI
- Run with Jupyter notebook
- Write a script



Managing computational pipelines

One-by-one

Good for alpha tests

Difficult to reproduce

Many opportunities for error

Jupyter notebook

Good when visualization/
manual inspection is necessary (beta tests)

Single-use

Doesn't scale

Scripting

Good when publishing/
distributing

Generic for multiple uses

Scales well

Managing computational pipelines

Scripting options

Shell script:

- Ex: https://github.com/zamanianlab/Filarid_IsoSeq/blob/master/main.sh
- Key features:
 - Can manage directories with variables
 - Checkpoint with comments (kind of)
 - Allows use of all command line tools, even Bash commands
 - No special language knowledge required
 - Could be made generic (but this one isn't)

Managing computational pipelines

Scripting options



Snakemake script:

- Key features:
 - Built-in directory and file management
 - Built-in checkpointing
 - Generic (with some restrictions)
 - Allows use of all command line tools, even Bash commands
 - Uses Python!
 - Uses conda!
 - Integrates with Slurm!

One machine, one project folder, one conda environment

.ipynb in VS Code

Cells, output inline



Python kernel

Runs cells in order



Shell session (bash or zsh)

Runs commands one at a time

Notebooks reach the shell
with ! or %%bash

One machine, one project folder, one conda environment

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Cells, output inline

Terminal in VS Code

Panel inside the editor

Terminal app

Same shell, no editor

Python kernel

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Shell script (.sh)
The same commands, saved

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Shell session (bash or zsh)

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Snakemake

Rules, DAG, checkpointing

Notebooks reach the shell
with ! or %%bash



Scripting pipelines in BIOL 343

- Each student will script their own pipelines
- Can choose which type you want to use
- Can use all available documentation/support online
- Can use your own conda envs

Live code activity...

1. Complete the snakemake tutorial on BOSE
 - a. Start a VS Code session
 - b. Install the snakemake extension
 - c. Start at Step 2 (above link) to setup the env, then go here and edit your snakefile in a VS Code session